DUNMAN SECONDARY SCHOOL

CANDIDATE NAME			
CLASS		**INDEX NUMBER**	

PRELIMINARY EXAMINATION 2024
SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

MATHEMATICS **4052/02**

Paper 2

21 August 2024

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Calculator Model

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.
If working is needed for any question it must be shown with the answer.
Omission of essential working will result in loss of marks.
The total of the marks for this paper is 90.

The use of an approved scientific calculator is expected, where appropriate.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π, use either your calculator value or 3.142, unless the question requires the answer in terms of π.

Total Marks

Mathematical Formulae

Compound interest

$$\text{Total amount} = P\left(1+\frac{r}{100}\right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi rl$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2}ab\sin C$$

$$\text{Arc length} = r\theta\text{, where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2}r^2\theta\text{, where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

Answer **all** the questions.

1 **(a)** Solve the inequality $\frac{2x+1}{2} \geq \frac{5-4x}{3}$.

Answer .... [2]

(b) Solve these simultaneous equations.

$$6x + 5y = 8$$
$$10x + 3y = 24$$

You must show your working.

Answer $x =$ ....

$y =$ .... [3]

(c) Express as a single fraction in its simplest form $\dfrac{3x}{(x-5)^2}+\dfrac{2}{5-x}$.

Answer .... [2]

(d) Simplify $\left(\dfrac{\sqrt[3]{p^2}}{\sqrt{2q}}\right)^{-6}$.

Answer .... [2]

(e) Simplify $\dfrac{36x^2-4}{9x^2-6x+1}$.

Answer .... [3]

2 The table below shows the distribution of the masses of durians harvested on Farm X.

Mass (m kg)	$0.5 \le m < 1$	$1 \le m < 1.5$	$1.5 \le m < 2$	$2 \le m < 2.5$	$2.5 \le m < 3$
Frequency	6	14	x	10	4

(a) Given that an estimate of the mean mass is 1.67 kg, find the value of x.

Answer $x =$ .... [2]

(b) Using the value of x in **(a)**, calculate an estimate of the standard deviation.

Answer .... [1]

(c) The mean mass of durians harvested on Farm Y is 1.45 kg.

Make a comment comparing the masses of the durians harvested on Farm X and on Farm Y.

....

....

.... [1]

(d) The masses of another 10 durians, each weighing between 1.5 kg to 2 kg, were added to the distribution for Farm X.

Explain how this change would have an effect on the standard deviation.

....

....

.... [1]

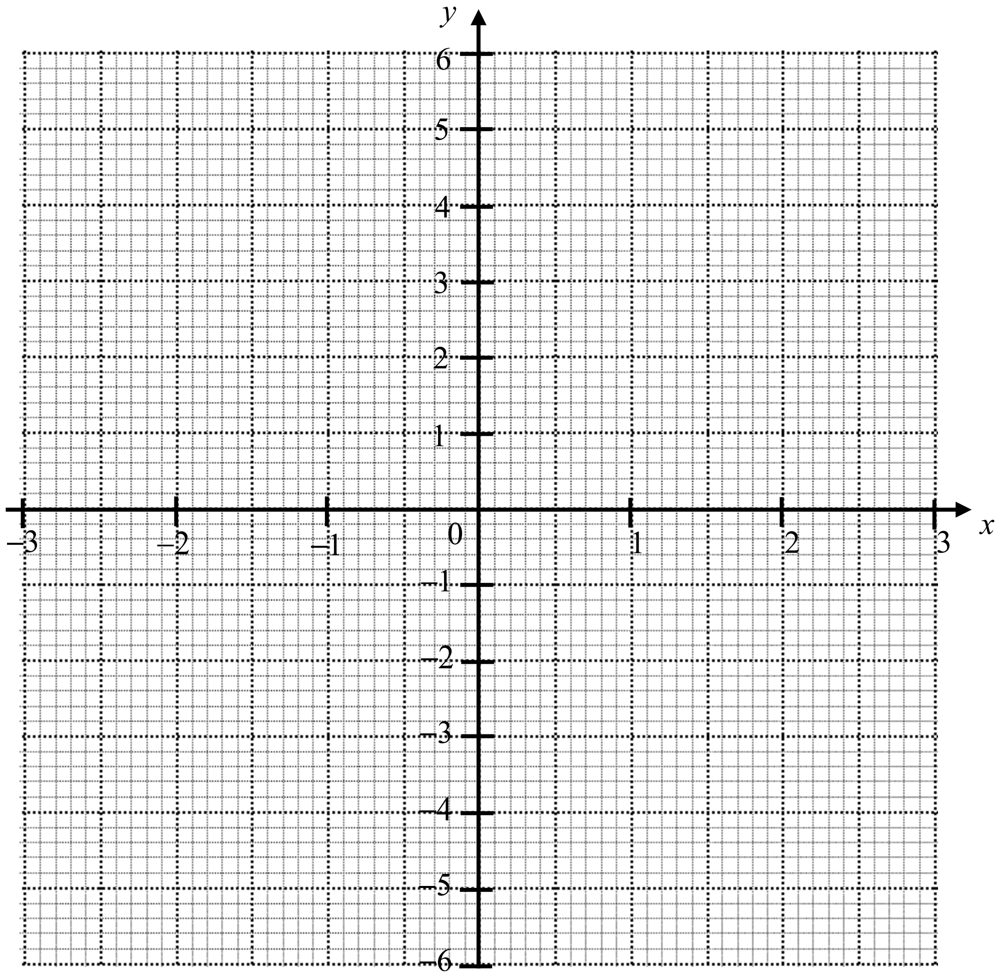
3 **(a)** Complete the table of values for $y = \frac{x^2}{4} - \frac{2}{x}$.

Values are given to 2 decimal places where appropriate.

x	–3	–2	–1	–0.5	0.5	1	2	3
y		2	2.25	4.06	–3.94	–1.75	0	1.58

[1]

(b) On the grid, draw the graph of $y = \frac{x^2}{4} - \frac{2}{x}$ for $-3 \le x \le 3$.



[3]

(c) By drawing a tangent, find the gradient of the curve at (–1, 2.25).

Answer .... [2]

(d) The equation $x^3 + 4x^2 + 12x - 8 = 0$ can be solved by finding the points of intersection of the straight line $y = ax + b$ and the curve $y = \frac{x^2}{4} - \frac{2}{x}$.

(i) Find the value of *a* and the value of *b*.

Answer $a =$ ……………………….

$b =$ ………………………. [2]

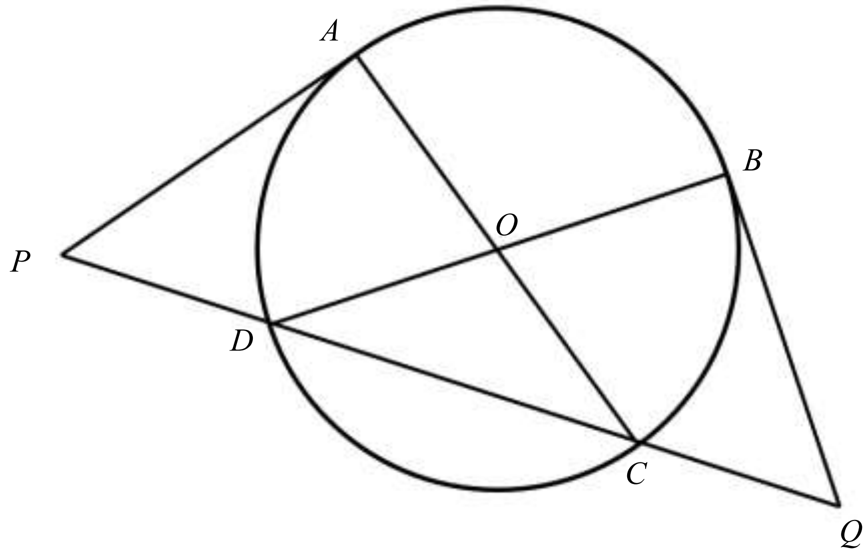
(ii) By drawing the line $y = ax + b$, solve the equation $x^3 + 4x^2 + 12x - 8 = 0$.

Answer $x =$ ………………………. [2]

(e) State the value of *k* for which $\frac{x^2}{4} - \frac{2}{x} = k$ has exactly 2 solutions.

Answer $k =$ ………………………. [1]

4 **(a)**



In the diagram, *AC* and *BD* are diameters of a circle, centre *O*.
AP and *BQ* are tangents to the circle.
PDCQ is a straight line.

(i) Show that triangle *APC* and triangle *BQD* are congruent.
Give a reason for each statement you make.

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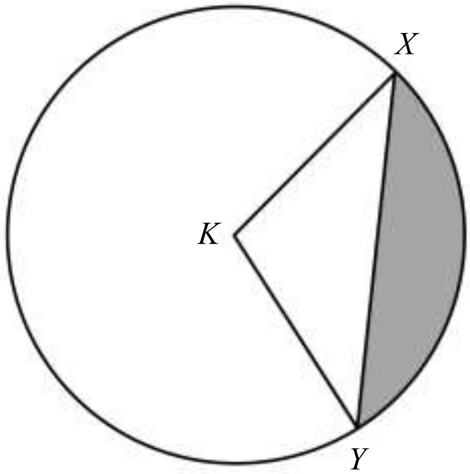
…………………………………………………………………………………… [3]

(ii) Given that angle *APD* is 54°, find angle *ABD*.
Give a reason for each step of your working.

Answer ……………………… ° [2]

(b)

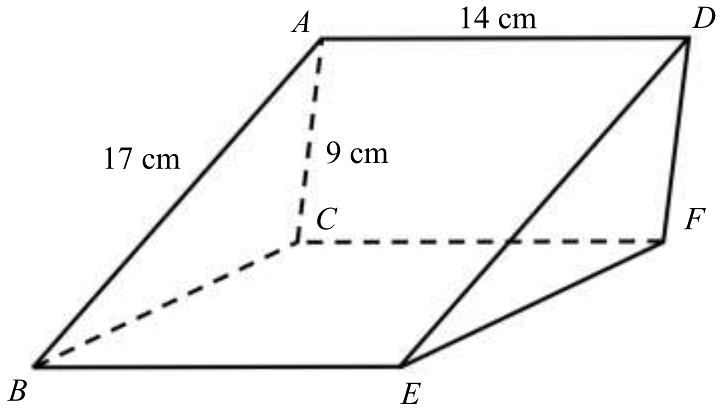


The diagram shows a circle, centre K, radius 10 cm.
Angle XKY = 1.8 radians.

Find the area of the **unshaded** section.

Answer ……………………… cm^2 [4]

5 The cross-section of the prism below is a triangle, where angle ACB is obtuse.
$AB = 17$ cm, $AC = 9$ cm, $AD = 14$ cm and angle $ABC = 25°$.



(a) Find obtuse angle ACB.

Answer ……………………… ° [2]

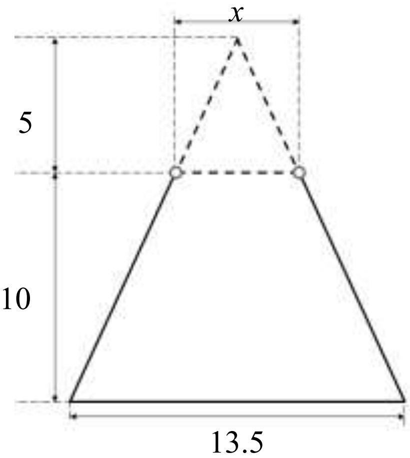
(b) Find the volume of the prism.

Answer ……………………… cm^3 [2]

(c) Find angle BFD.

Answer ……………………… ° [5]

6



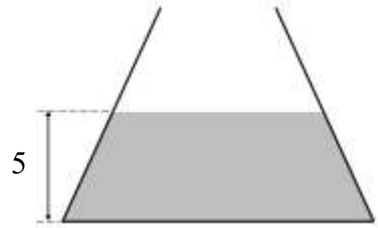
A conical flask is made by removing a small, similar cone from a larger cone.
The height of the flask is 10 cm and the height of the small cone is 5 cm.
The diameter of the bottom of the flask is 13.5 cm.
The thickness of the flask is negligible.

(a) The diameter of the top of the flask is x cm.

Find the value of x.

Answer x = ……………………… [2]

(b)



Cathy pours water into the flask.
The surface of the water is 5 cm from the bottom of the flask.

(i) Cathy thinks that the flask is filled to 50% of its total capacity

Explain why she is wrong.

……………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(ii) Calculate the volume of water in the flask.

Answer ……………………… cm^3 [4]

(iii) Cathy pours the water from this flask into a second container.
The second container is in the shape of a cuboid with a square base.
The depth of water in the cuboid is 7 cm.

Calculate the length of the square base of the cuboid.

Answer ……………………… cm [2]

7 At a theme park, the price of one ticket on a weekday is \$80 for adults, \$30 cheaper for children and \$20 cheaper for seniors.
At a theme park, the price of one ticket on a day at the weekend is \$110 for adults, \$40 cheaper for children and \$25 cheaper for seniors.

This information can be represented by the matrix $\mathbf{T} = \begin{pmatrix} 80 & 110 \\ -30 & -40 \\ -20 & -25 \end{pmatrix}$.

Two groups, A and B, want to spend a weekday and a day at the weekend at the theme park.

(a) Group A wants to buy 30 tickets, of which 7 are for children and 5 are for seniors.
Group B wants to buy 50 tickets, of which 12 are for children and 8 are for seniors.

Represent this information in a 2×3 matrix, **P**.

Answer **P** = [1]

(b) Evaluate the matrix $\mathbf{R} = \mathbf{PT}$.

Answer **R** = [2]

(c) Explain what the first element of matrix **R** represents.

……………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(d) Group B receives a discount of 10% for all ticket prices.

Calculate the total cost of the tickets for Group B.

Answer $ .... [2]

(e) The elements of the matrix **S**, where **S** = **KT**, represent the actual prices of each ticket for adults, children and seniors on a weekday and a day at the weekend respectively.

Write down the matrix **K**.

Answer **K** = [1]

8 *PQRS* is a rectangle.
P is the point (3, –1) and *R* is the point (2, 7).
$\overrightarrow{PQ} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$.

(a) Find the length of the line *PQ*.

Answer .... units [2]

(b) Find the equation of the line *RS*.

Answer .... [4]

(c) A circle, with centre C, can be drawn such that the points P, Q, R and S lie on its circumference.

(i) Explain why it is possible for the circle, with centre C, to be drawn such that the points P, Q, R and S lie on its circumference.

……………………………………………………………………………………………

……………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(ii) Find $\overrightarrow{PC}$.

Answer $\overrightarrow{PC} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [2]

(iii) Find the position vector of C.

Answer $\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

9 A bag contains 7 green marbles, 1 red marble and x black marbles.
Two marbles are drawn from the bag, without replacement.

(a) Find, in terms of x, the probability that the first marble drawn is red.

Answer …………………… [1]

(b) The probability that both marbles are of the same colour is $\frac{2}{5}$.

Write down an equation to represent this information and show that it simplifies to

$$3x^2 - 35x + 98 = 0 \ .$$

Answer

[3]

(c) Solve the equation $3x^2 - 35x + 98 = 0$.

Answer $x =$ ………………… or ………………… [2]

(d) Find, as a fraction in its simplest form, the probability that a black marble and a red marble is drawn.

Answer …………………… [2]

10 The Total Daily Energy Expenditure (TDEE) is an estimation of how many calories a person uses per day and can be calculated by:

$$TDEE = [10 \times weight\ \text{(kg)} + 6.25 \times height\ \text{(cm)} - 5 \times age\ \text{(years)} + 5] \times Activity\ Level$$

The Activity Level can be determined by the table below.

Activity Level	**Description**
1.2	Sedentary: 1 to 3 sessions of exercise a month
1.375	Light: 1 to 3 sessions of exercise a week
1.55	Moderate: 4 to 5 sessions of exercise a week
1.725	High: 6 to 7 sessions of exercise a week
1.9	Extreme: 2 sessions of exercise a day

(a) Find the TDEE of a 25 year old person who is 172 cm tall, weighs 72 kg and has 6 sessions of exercise a week.

Answer ………………………… calories/day [1]

Additional calories are used when a person performs certain exercises and can be calculated by:

$$Calories = MET \times weight\ \text{(kg)} \times duration\ \text{(hours)}\ ,$$

where Metabolic Equivalent of Task (MET) can be determined by the table below.

MET	**Type of exercise**
3.1	Walking
7.3	Badminton
7.8	Swimming
8.8	Jogging
10.3	Basketball

(b) Find the additional calories used by a person who weighs 65 kg and plays badminton for 45 minutes.

Answer ………………………… calories [2]

Stan is 42 years old, weighs 75 kg and is 168 cm tall.
He aims to lose some weight.
He learns that he needs to use about 3850 calories more than the calories that he eats every week.
He decides to eat about 2150 calories every day.
He has a one-hour session of swimming every week and wants to add 3 sessions of jogging to his weekly schedule.

(c) Suggest a reasonable duration for Stan to jog during each session.
Justify any decision you make and show your calculations clearly.

Answer

………

……… [7]

END OF PAPER

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